

Practical 7. String and File Operation

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In this practical, you will write python code for three problems. You can ask the instructors for help. You may also need to look for online references to complete this assignment, which is good practice on how to ask right questions. Please don't be shy about asking

You are expected to use your GitHub account from previous weeks to store your code. Please create a directory called Practical 7 in your GitHub repository and put all your python scripts from this practical there.

1 Learning objectives

- Create and manipulate strings in python
- Search and find regular expression patterns in strings in Python
- Read and write files in Python

2 Open Reading Frames (ORF)

An mRNA molecule is translated into a protein using a triplet code, where each group of three nucleotides represents one amino acid. Some triplets are specifically used to indicate the start and end of the protein sequence. The complete protein-coding sequence within an mRNA is known as the open reading frame (ORF). The start codon 'AUG' indicates the beginning of the ORF in the mRNA sequence, while the stop codons 'UAA', 'UAG', and 'UGA' signal the end of the ORD.

Today we will write code to identify the longest potential open reading in within gene sequences. Genes with multiple start and stop codons sites can generate open reading frames of variable lengths.

Please create a new python script **largest_orf.py** and create a string variable seq:

```
seq = 'AAGAUACAUGCAAGUGGUGUGUCUGUUCUGAGAGGGCCUAAAAG'
```

You should now write code which will identify the largest ORF that can be generated from this sequence and report the length of that ORF in nucleotides. Note that the start codon can begin at any position within the string variable, it does not have to be at the very beginning or within a multiple of three nucleotides from the beginning.

3 Stop codon usage

A common format to store sequence data is FASTA. In a fasta/fa file, each sequence starts with a ">" symbol followed by the name of the sequence and the description of the sequence and the remaining line(s) contain the sequence itself. Here is a fasta file downloaded from ftp://ftp.ensemblgenomes.org/pub/fungi/release-46/fasta/saccharomyces_cerevisiae/cdna/. The file contains the cDNA sequences of all the genes in yeast *Saccharomyces Cerevisiae*. You can view the first lines of the file by using the **head** command in your terminal. The name line has a list of gene information, such as the position in the chromosome, the gene name, the biotype, etc...

3. Generates a well-labelled pie chart showing the distribution of all in-frame codons produced by genes contain the user-specified stop codon. This pie chart should be printed out to a file, not just displayed on the screen.

5 For your portfolio

largest_orf.py

- The marker will check that the script contains the variable `seq` shown above and prints out the longest ORF and its length.
- The marker will check that these output change when the variable `seq` is modified.

stop_codons.py

- The marker will check that the script can read the original fasta file and create a new file **stop_genes.fa**.
- The sequence name should contain only the gene name and which of the three stop codons are present at least once in the gene sequence.

count_codons.py

- The marker will check that the script correctly accepts one of the three stop codons and only quantifies the codon usage across genes containing that stop codon.
- The marker will verify that a well-labelled pie chart is created and that the results of this are updated when a new stop codon is provided.

You can add or edit things after the Practical and Tutorials have finished. We do not look at the commit date, we just want it all to be there!